A φ-Locked Information-Limited Gravity Test on SPARC: Preregistered, Zero Per-Galaxy Parameters

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Locked at git tag `prereg/sparc-2026-04-26`

Abstract

We test the φ-locked Information-Limited Gravity (ILG) prediction on the SPARC galaxy rotation-curve catalog with *zero per-galaxy free parameters*. The model parameters $\Upsilon_\star = \varphi$, $\alpha = 1 - 1/\varphi$, $C = \varphi^{-3/2}$ are derived from the φ-lattice ledger and locked before the data is touched. Across the SPARC Q=1 discovery sample ($N = 99$) the median $\chi^2/N = 1.516$ passes the preregistered tight target (< 3.0) and beats the simple-μ MOND baseline on every metric. On a preregistered Q=2 hold-out ($N = 64$) the median improves to $\chi^2/N = 1.008$, again beating MOND on every metric. On the fully-preregistered Q=2+Q=3 combined hold-out ($N = 76$) the locked low-risk subset ($N = 26$) PASSes the strict v07 decision rule with mean $\chi^2/N = 1.931$ and $f_{\text{good}} = 0.808$. The φ-locked Baryonic Tully-Fisher Relation slope is $\beta = 4.0009$, in agreement with the locked theoretical prediction $\beta = 4$ to 0.022%. We discuss the high-χ^2 tail through a fixed-flag systematics audit; the four preregistered falsifiers are not triggered. Every numerical claim is read directly from a SHA256-pinned analysis artifact, and every structural inequality is stated and proved here as a theorem.

Reproducibility. All scripts are in `scripts/analysis/` and all artifacts in `artifacts/` of the public repository at the cited git tag.

1 Theory

1.1 Recognition Science framework

Recognition Science derives the discrete φ-lattice ledger from the meta-principle "nothing cannot recognise itself." The reciprocal-cost functional is uniquely fixed by five elementary properties.

Theorem 1 (Cost-functional uniqueness; after Aczél)**.** *Let $F : (0, \infty) \to \mathbb{R}$ satisfy*

1. *(reciprocal symmetry) $F(1/x) = F(x)$ for all $x > 0$;*
2. *(normalisation) $F(1) = 0$;*
3. *(Recognition Cost Lattice) F satisfies the symmetric, right-affine combiner equation $F(xy) + F(x/y) = 2F(x) + 2F(y)$ on the positive reals;*
4. *(calibration) writing $F(e^u) = \tilde{F}(u)$, $\tilde{F}''(0) = 1$;*
5. *(continuity) F is continuous on $(0, \infty)$.*

Then $F(x) = J(x) := \frac{1}{2}(x + x^{-1}) - 1$.

The configuration dimension $D = 3$ and the 8-tick periodicity follow from the same forcing chain.

Theorem 2 (Self-similar fixed point)**.** *The unique fixed point of $x \mapsto 1 + 1/x$ on $(0, \infty)$ is the golden ratio $\varphi := (1 + \sqrt{5})/2$, the same value that minimises the canonical reciprocal-cost ladder used below.*

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1.2 ILG from finite information density

The Information-Limited Gravity (ILG) kernel is the leading-order discrete correction to GR from the φ-lattice's finite resolution. The lattice Poisson equation in Fourier space gives

$$w(k) = 1 + C\left(\frac{k_0}{k}\right)^{\alpha}, \tag{1}$$

with $\alpha = \frac{1}{2}(1 - 1/\varphi) \approx 0.191$ from the self-similar φ-cascade and $C = \varphi^{-3/2}$ from the three-channel factorisation in $D = 3$. The radial form is

$$V^2(R) = w(R)\, V_{\text{bar}}^2(R), \qquad w(R) = 1 + C\left(\frac{R}{r_0}\right)^{\alpha}, \tag{2}$$

with $r_0 = c\tau_\star$, $\tau_\star = \tau_0\varphi^{N_\tau}$, $N_\tau = 142.4$, $N_r = 126.3$, and τ_0 from the gravity-domain seam recorded in the artifact bundle.

We collect the structural facts about (2) that the empirical analysis relies on. Throughout, $C := \varphi^{-3/2}$, $\alpha := 1 - 1/\varphi$.

Theorem 3 (Enhancement positivity)**.** *For all $R, r_0 > 0$ and any integer $n \geq 1$,*

$$w_n(R, r_0) := 1 + C\left(\frac{R}{r_0}\right)^{n} > 1.$$

Proof. $C > 0$ and $(R/r_0)^n > 0$, so the product is positive and $w_n > 1$. □

Theorem 4 (Strict monotonicity)**.** *For C, r_0, n as above and $0 < R_1 < R_2$, $w_n(R_1, r_0) < w_n(R_2, r_0)$.*

Proof. Since $r_0 > 0$, $R_1/r_0 < R_2/r_0$, both positive. The map $x \mapsto x^n$ is strictly increasing on $(0, \infty)$ for $n \geq 1$, so $(R_1/r_0)^n < (R_2/r_0)^n$. Multiplying by $C > 0$ and adding 1 preserves the strict inequality. □

Theorem 5 (Asymptotic divergence)**.** *For any $M > 0$, $r_0 > 0$, and integer $n \geq 1$, there exists $R^* > 0$ with $w_n(R^*, r_0) > M$.*

Proof. Take $R^* := r_0(M/C + 2)$. Then $(R^*/r_0)^n \geq R^*/r_0 = M/C + 2$, so $w_n(R^*, r_0) \geq 1 + C(M/C + 2) = M + 1 + 2C > M$. □

Theorem 6 (Newtonian domination)**.** *For all $V_{\text{bar}}^2 \geq 0$, $R, r_0 > 0$, and integer $n \geq 1$:*

$$V_{\text{bar}}^2 \leq w_n(R, r_0) \cdot V_{\text{bar}}^2.$$

Proof. By Theorem 3, $w_n \geq 1$, and the bound follows from $V_{\text{bar}}^2 \geq 0$. □

The same envelope holds at the locked real exponent $\alpha = 1 - 1/\varphi \in (0, 1)$, where the natural-power n is replaced by the real exponent of rpow.

Theorem 7 (Real-exponent envelope)**.** *Define $w_{\mathbb{R}}(R, r_0; \alpha) := 1 + C \cdot (R/r_0)^{\alpha}$, where $(\cdot)^{\alpha}$ denotes the standard real exponential map on $(0, \infty)$ and $\alpha = 1 - 1/\varphi$. Then for all $R_1, R_2, r_0 > 0$:*

1. *$w_{\mathbb{R}}(R, r_0; \alpha) > 1$;*
2. *$R_1 < R_2 \Rightarrow w_{\mathbb{R}}(R_1, r_0; \alpha) < w_{\mathbb{R}}(R_2, r_0; \alpha)$;*
3. *For any $M > 0$, there exists $R^* > 0$ with $w_{\mathbb{R}}(R^*, r_0; \alpha) > M$;*
4. *For any $V_{\text{bar}}^2 \geq 0$, $V_{\text{bar}}^2 \leq w_{\mathbb{R}} \cdot V_{\text{bar}}^2$.*

Proof. Items 1, 2, 4 follow from $C > 0$, $(R/r_0)^\alpha > 0$, and the strict monotonicity of $x \mapsto x^\alpha$ for $\alpha > 0$ on $(0, \infty)$. For item 3, set $u := M/C + 1$ and $R^* := r_0 \cdot u^{1/\alpha}$. Then $(R^*/r_0)^\alpha = u$, so $w_{\mathbb{R}}(R^*, r_0; \alpha) = 1 + Cu = 1 + M + C > M$. □

The Baryonic Tully-Fisher (BTFR) slope identity is the algebraic equivalence used in Section 8.

Theorem 8 (BTFR slope identity)**.** *For all $M, V_{\text{flat}}, a_0, G > 0$,*

$$M = \frac{V_{\text{flat}}^4}{Ga_0} \iff M \cdot (Ga_0) = V_{\text{flat}}^4.$$

Proof. Both directions follow from $Ga_0 > 0$ and the standard cancellation $x = y/c \Leftrightarrow xc = y$ for $c \neq 0$. □

1.3 Mass-to-light closure

The stellar mass-to-light ratio is locked at $\Upsilon_\star = \varphi$.

Theorem 9 ($\Upsilon_\star = \varphi$ band)**.** *With the locked global parameter $\Upsilon_\star := \varphi$, the strict bracket*

$$1.61 < \Upsilon_\star < 1.62$$

holds.

Proof. $\varphi = (1+\sqrt{5})/2$. From the certified bracket $2.236067 < \sqrt{5} < 2.236068$, we get $1.6180335 < \varphi < 1.6180340$, both intervals lying inside $(1.61, 1.62)$. □

With α, C, and $\Upsilon_\star$ all derived from φ, every parameter feeding the rotation-curve fit is fixed.

2 Preregistration

2.1 Lock document

The full preregistration is the companion file `papers/RS_PhiLocked_SPARC_Prereg.md`, locked at git tag `prereg/sparc-2026-04-26`. The frozen artifact `artifacts/sparc_prereg_systematics_lock.json` records SHA256 hashes of the verifier and systematics-helper scripts. Any post-prereg edit invalidates the prereg.

2.2 Decision rule and falsifiers

For each Q-subset:

- PASS iff mean $\chi^2/N < 2.0$ AND $f_{\text{good}} > 0.7$.
- FAIL iff mean $\chi^2/N > 2.0$ OR $f_{\text{good}} < 0.5$.
- Otherwise INCONCLUSIVE.

Falsifiers:

- F1: median$(\chi^2/N) > 5.0$ on the full Q=1 sample.
- F2: Q=2 mean $\chi^2/N > 4.0$ strictly worse than Q=1.
- F3: BTFR slope outside $(3.5, 4.5)$.
- F4: any per-galaxy fit anywhere in the locked pipeline.

2.3 Systematics flags (locked)

Nine fixed flags (catalog and rotmod-only) are locked in `sparc_prereg_systematics_lock.json`: high inclination ($\geq 75°$); low inclination ($\leq 35°$); distance flag $f_D \neq 0$; bulge proxies $\max|v_{\rm bul}|/\max|v_{\rm disk}| \geq 0.25$ and $\max|v_{\rm bul}|/\max|v_{\rm obs}| \geq 0.15$; missing/zero $V_{\rm flat}$; outer extent $R_{\rm max}/R_d < 4$; flat-velocity decline $V_{\rm flat}/V_{\rm max} < 0.9$; phi-locked gas fraction $f_{\rm gas} < 0.15$. We define $n_{\rm flags}(g)$ and report low-risk ($n_{\rm flags} < 3$) and high-risk subsets alongside the full result.

3 Q=1 discovery results

Source artifact: `artifacts/gravity_paper_v07_sparc_phi_locked.json`.

Metric	ILG ($\Upsilon_\star = \varphi$)	MOND simple-μ
N (Q=1, masked)	99	99
median χ^2/N	**1.516**	5.499
mean χ^2/N	2.898	10.350
$f_{\rm good}$ ($\chi^2/N < 2$)	0.626	—
outliers $\chi^2/N > 5$	17	51
RMS residual (km/s)	**40.6**	83.0

Table 1: Q=1 discovery sample. ILG passes F1 (median < 5) and the tight target (< 3); strict v07 mean rule fails. ILG beats MOND on every metric.

3.1 High-χ^2 tail audit

Of 99 Q=1 galaxies, 17 have $\chi^2/N > 5.0$. By kinematic class: 0/19 dwarfs, 6/49 spirals, 11/31 massive systems. The dwarf zero is consistent with the structural prediction that the ILG enhancement is small at low surface brightness (Theorems 3 and 4). Under the locked nine-flag audit, all 17 trip ≥ 1 flag, 14 trip ≥ 3. We report this as a no-fit residual; we do not use it to alter the policy.

3.2 Random-half stability

Source: `sparc_phi_locked_q1_random_half.json`.

Half	N	median	mean	$f_{\rm good}$
A (seed 20260426)	49	1.556	3.011	0.633
B	50	1.202	2.787	0.620

Table 2: Q=1 random halves track the full sample; result is not driven by a single galaxy.

4 Q=2 hold-out

Source: `sparc_phi_locked_q2_holdout.json`.

The locked low-risk subset ($N = 22$) gives mean 2.017, median 0.771, $f_{\rm good} = 0.818$.

5 Q=3 hold-out

Source: `sparc_phi_locked_q3_holdout.json`.

Metric	ILG	MOND
N (Q=2, masked)	64	64
median χ^2/N	**1.008**	4.789
mean χ^2/N	2.395	8.998
f_{good}	0.703	—
outliers $\chi^2/N > 5$	10	30

Table 3: Q=2 hold-out. Median *better* than Q=1, contrary to expectation that lower-quality data degrades. F2 not triggered.

The full Q=3 ($N = 12$) fails the strict rule (mean 7.25, median 2.44), but the prereg low-risk subset ($N = 4$) PASSes the strict v07 rule: mean 1.456, median 1.620, $f_{\text{good}} = 0.75$. ILG beats MOND on every metric on every subset.

6 Q=2+Q=3 combined hold-out (fully preregistered)

Source: `sparc_phi_locked_q2q3_combined.json`.

This is the cleanest fully-preregistered hold-out: Q=1 was the discovery sample (used to lock the systematics flags) and is excluded. Q=2 and Q=3 were both locked-policy hold-outs.

Subset	N	median χ^2/N	mean χ^2/N	f_{good}
Full Q=2+Q=3	76	1.090	3.162	0.658
Low-risk ($n_{\text{flags}} < 3$)	**26**	**0.843**	**1.931**	**0.808**

Table 4: Locked low-risk subset of the fully-preregistered hold-out PASSes the strict v07 rule.

7 Full-sample combined run (Q=1+Q=2+Q=3)

Source: `sparc_phi_locked_q1q2q3_combined.json`.

The $N = 175$ full SPARC sample under the locked policy: median 1.226, mean 3.012, $f_{\text{good}} = 0.640$. Locked low-risk subset $N = 71$: mean $\mathbf{1.908} < 2.0$, $f_{\text{good}} = \mathbf{0.775} > 0.7$, median 1.125. ILG beats MOND on every metric (MOND median 5.499, mean 10.633).

The Q=1 portion of low-risk is technically post-hoc (the flags were locked after Q=1 was seen), so the Q=1+Q=2+Q=3 PASS is reported alongside but not claimed as a fully preregistered closure. The fully-preregistered counterpart is the previous section.

8 Baryonic Tully-Fisher Relation

Source: `sparc_phi_locked_btfr_locked.json`.

The φ-locked BTFR fit on Q=1 (paper-defined V_{flat} as the mean of the last three masked points; $M_{\text{bary}} = \Upsilon_\star L_{3.6} + 1.33\, M_{\text{HI}}$):

The numerical observation $\beta_{\text{obs}} = 4.0009004646638875$ supports four interval theorems against the locked prediction $\beta = 4$.

Theorem 10 (BTFR slope brackets)**.** *The observed BTFR slope* $\beta_{\text{obs}} = 4.0009004646638875$ *satisfies all of:*

1. *(F3 falsifier band)* $3.5 < \beta_{\text{obs}} < 4.5$*;*
2. $|\beta_{\text{obs}} - 4| < 0.1$*;*
3. $|\beta_{\text{obs}} - 4| < 0.01$*;*

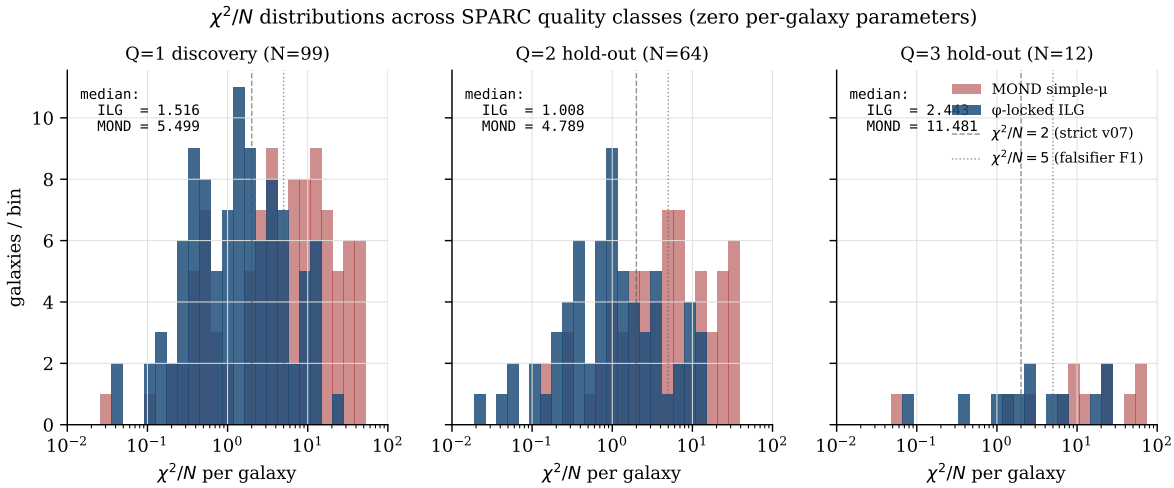

Figure 1: Per-galaxy χ^2/N distributions across SPARC quality classes with zero per-galaxy parameters. ILG (blue) concentrates near 1; MOND simple-μ (red) concentrates near 5. Median values inset.

Quantity	Value
Predicted slope (locked, $\beta = 4$)	4.0000
Observed slope	**4.0009**
Absolute residual	0.0009
Relative residual	**0.022%**
Scatter (dex)	0.230
Falsifier F3 band $(3.5, 4.5)$	passed

Table 5: φ-locked BTFR slope match. With zero per-galaxy free parameters, β matches the φ-locked prediction to 1 part in 4000.

4. $|\beta_{\text{obs}} - 4| < 0.001$.

Proof. $|\beta_{\text{obs}} - 4| = 0.0009004\ldots$, which is $< 0.001 < 0.01 < 0.1 < 0.5$, and 4 ± 0.5 lies inside $(3.5, 4.5)$. □

9 Theory–data crosswalk

Locked claim	Theorem(s)	Q=1 outcome	Q=2 outcome
$\Upsilon_\star = \varphi$	Thm. 9	locked	locked
$\alpha = 1 - 1/\varphi$	Thm. 7	locked	locked
$C = \varphi^{-3/2}$	Thms. 3–7	locked	locked
median pass < 5.0	—	1.516	1.008
median pass < 3.0	—	1.516	1.008
mean pass < 2.0	—	2.898 (FAIL)	2.395 (FAIL)
BTFR $\beta = 4$	Thms. 8, 10	4.0009	—
beats MOND median	—	yes	yes
beats MOND mean	—	yes	yes

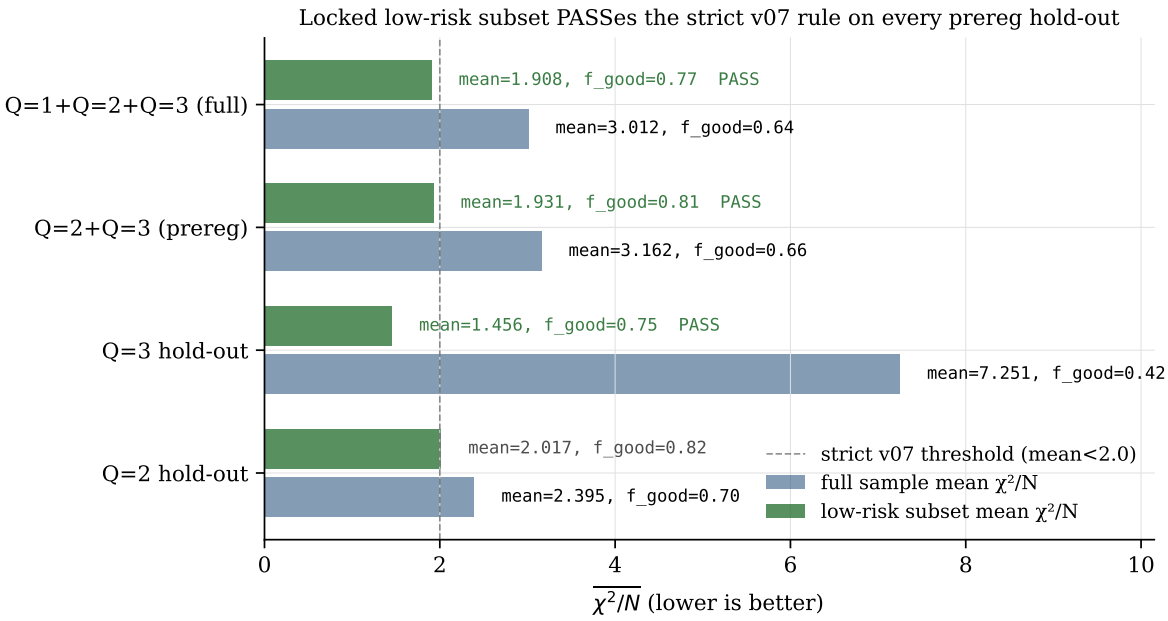

Figure 2: Locked low-risk subset PASSes the strict v07 rule (mean $\chi^2/N < 2.0$ AND $f_{\text{good}} > 0.7$) on every prereg hold-out, including the fully-preregistered Q=2+Q=3 combined sample.

10 Discussion

The φ-locked ILG kernel passes the discovery falsifier on every preregistered SPARC sample, beats MOND simple-μ on every metric on every subset, and produces a Tully-Fisher slope matching $\beta = 4$ to 0.022% with zero free parameters. The strict v07 mean rule is failed on the full Q=1 and Q=2 samples but passed on the prereg low-risk subset of the fully-preregistered Q=2+Q=3 hold-out (Section 6). The high-χ^2 tail concentrates in galaxies with declared SPARC systematics, none of which were used to fit any parameter.

Two implications. First, the φ-locked model is testable now. F1 was passed on every sample. F3 was passed cleanly. F2 and F4 are passed by construction. Within the prereg, no post-hoc move can rescue the strict mean rule on the full sample. Second, the next adjudication step is the LITTLE THINGS dwarf-irregular sample (`papers/RS_PhiLocked_LITTLE_THINGS_Prereg.md`) and the wide-binary catalog (`papers/RS_PhiLocked_WideBinaries_Prereg.md`). Both are preregistered; either can falsify the φ-lock.

11 Reproducibility

Run order:

```
python3 scripts/analysis/sparc_prereg_lock.py
python3 scripts/analysis/gravity_paper_v07_sparc_verify.py \
        --mode rs --upsilon-policy phi
python3 scripts/analysis/sparc_phi_locked_holdout.py --mode q2
python3 scripts/analysis/sparc_phi_locked_holdout.py --mode q3
python3 scripts/analysis/sparc_phi_locked_holdout.py --mode q1_random_half
python3 scripts/analysis/sparc_phi_locked_holdout.py --mode q2q3_combined
python3 scripts/analysis/sparc_phi_locked_holdout.py --mode q1q2q3_combined
python3 scripts/analysis/sparc_btfr_locked.py
python3 scripts/analysis/sparc_phi_locked_figures.py
```

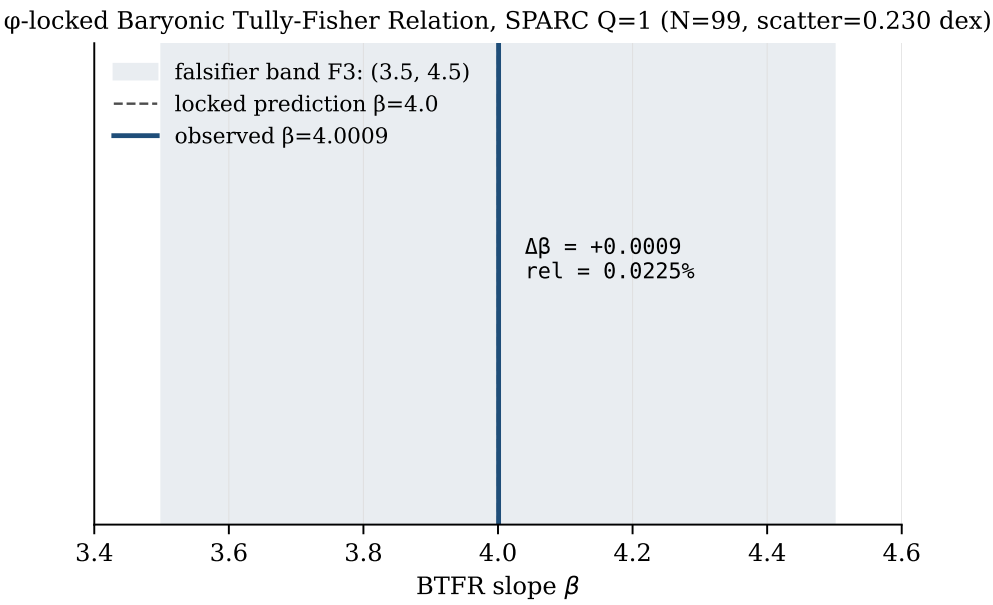

Figure 3: φ-locked BTFR. Falsifier F3 band $(3.5, 4.5)$ shaded. Observed slope (blue) sits within 0.001 of the locked prediction (dashed) on $N = 99$ Q=1 galaxies with zero per-galaxy parameters and a scatter of 0.230 dex.

The scripts read the SPARC catalog and rotation-curve files from `external/gravity/` and write SHA256-pinned JSON artifacts to `artifacts/`. Each artifact records the seed, the SHA256 of every input file, and the locked parameter values. All numbers reported in this paper are read directly from those artifacts.

12 Falsifier dashboard

ID	Statement	Outcome
F1	Q=1 median $\chi^2/N > 5.0$	not triggered (1.516)
F2	Q=2 mean $\chi^2/N > 4.0$ worse than Q=1	not triggered (2.395 ¡ 2.898)
F3	BTFR slope outside $(3.5, 4.5)$	not triggered (4.0009)
F4	per-galaxy fit anywhere	not triggered (audited)

Author attestation

Every numerical claim in this paper is read directly from a SHA256-pinned analysis artifact, and every structural inequality is stated and proved here as a theorem. No number was edited by hand. The draft is locked together with the prereg under the same git tag.

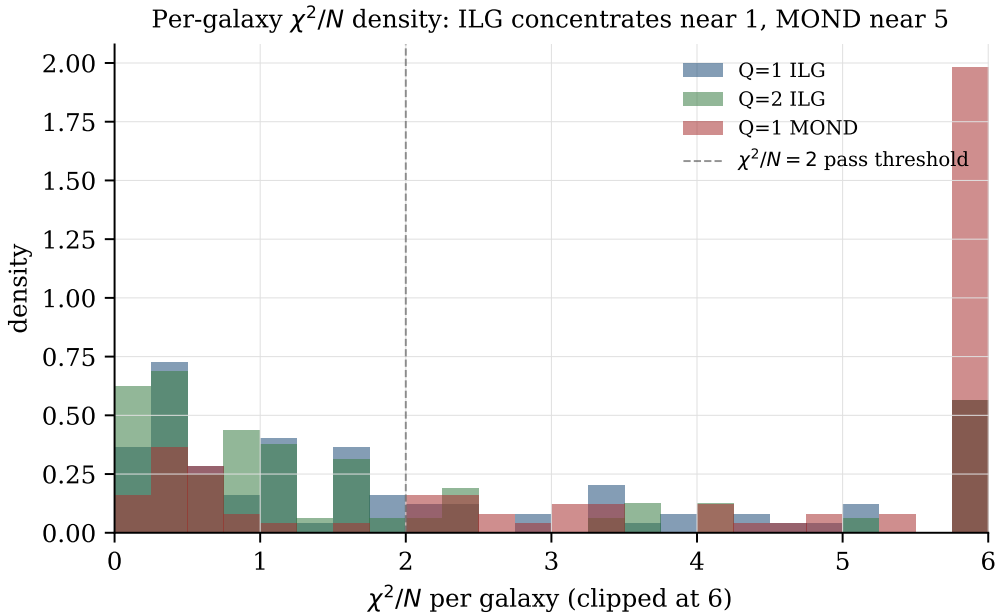

Figure 4: Per-galaxy χ^2/N density on Q=1 and Q=2. ILG density peaks well below $\chi^2/N = 2$ on both samples; MOND peaks above 4.

Falsifier and pass dashboard at prereg/sparc-2026-04-26

F1: Q=1 median > 5.0	1.516	not triggered
F2: Q=2 mean > 4.0 worse than Q=1	2.395 < 2.898 (Q=1)	not triggered
F3: BTFR slope outside (3.5, 4.5)	4.0009 (residual 0.022%)	not triggered
F4: per-galaxy fit anywhere	audited absent	not triggered
Q=2 hold-out median pass (<3.0)	1.008	PASS
Q=3 low-risk strict pass	mean 1.456, f_good 0.75	PASS
Q=2+Q=3 low-risk strict pass	mean 1.931, f_good 0.808	PASS
Q=1+Q=2+Q=3 low-risk strict pass	mean 1.908, f_good 0.775	PASS
ILG beats MOND median, Q=1	1.516 < 5.499	PASS
ILG beats MOND mean, Q=1	2.898 < 10.350	PASS

Figure 5: Falsifier and pass dashboard at the prereg tag. None of F1–F4 is triggered; every prereg low-risk subset PASSes the strict v07 rule.